



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004
TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All *necessary* working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Hand in your answer booklets in 5 sections.
Section A (Questions 1 & 2),
Section B (Questions 3 & 4),
Section C (Questions 5 & 6),
Section D (Questions 7 & 8) and
Section E (Questions 9 & 10).
- Start each **NEW** section in a separate answer booklet.

Total Marks - 120 Marks

- Attempt Sections A - E
- All questions are of equal value.

Examiner: *P. Parker*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Total marks – 120
Attempt Questions 1 – 10
All questions are of equal value

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

SECTION A (Use a SEPARATE writing booklet)

Question 1 (12 marks)		Marks
(a)	Evaluate $\log_e \left(\tan \frac{5\pi}{12} \right)$ leaving your answer correct to 3 significant figures	2
(b)	Differentiate $\sqrt{5x}$	2
(c)	Solve $2t^2 - t - 15 = 0$	2
(d)	Find a primitive of $3 - 2x$	2
(e)	Solve the pair of simultaneous equations $y = 2x$ $3x + 2y = 14$	2
(f)	Solve $3 - 4x < 1$ and graph the solution on a number line	2

Question 2 (12 marks)

Marks

(a) Differentiate

(i) $(1 + \cos 2x)^3$ 2

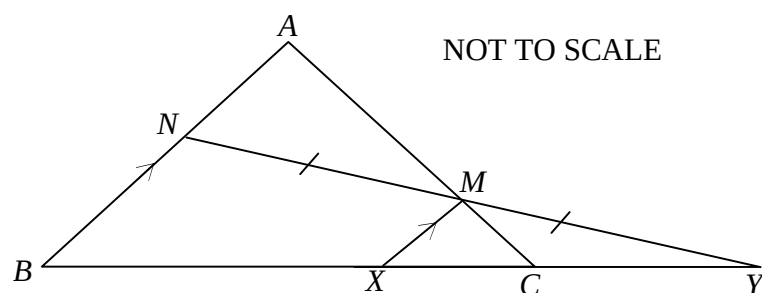
(ii) $x^2 e^{x+2}$ 2

(b) Find:

(i) $\int \frac{\cos x}{\sin x} dx$ 1

(ii) $\int_{\frac{1}{2}}^2 \left(1 - \frac{1}{x^2}\right) dx$ 2

(c)



In the diagram above $\triangle ABC$ is isosceles, with $AB = AC$.
 M is the midpoint of the line NY and $XM \parallel AB$.

(i) By using similar triangles, or otherwise, show that $\frac{MX}{NB} = \frac{1}{2}$ 2

(ii) Hence show that $\frac{MC}{NB} = \frac{1}{2}$ 1

(d) The graph of $y = g(x)$ passes through the point $(2, 4)$ and $g'(x) = 4 - 3x^2$. 2

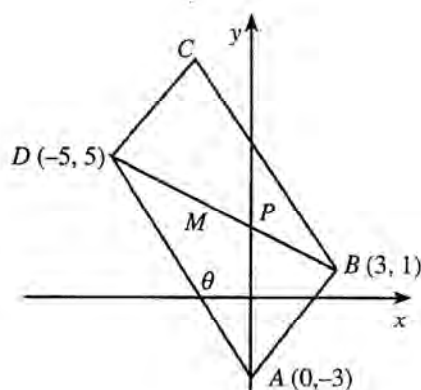
Find $g(x)$.

SECTION B (Use a SEPARATE writing booklet)

Question 3 (12 marks)

Marks

In the diagram below A , B and D have coordinates $(0, -3)$, $(3, 1)$ and $(-5, 5)$ respectively.
The angle θ is the angle the line AD makes with the positive direction of the x axis.



- (i) Find the gradient of the line AD .
Hence find θ to the nearest degree. 2
- (ii) Find the coordinates of M , the midpoint of BD . 1
- (iii) Find the coordinates of C , so that $ABCD$ is a parallelogram. 1
- (iv) Show that the line AB has equation $4x - 3y - 9 = 0$. 2
- (v) Find the perpendicular distance between D and AB . 1
- (vi) Find the area of parallelogram $ABCD$. 2
- (vii) The line BD has equation $x + 2y - 5 = 0$ and meets the y axis at P .
Write down the three inequalities that define the region inside $\triangle ABP$. 3

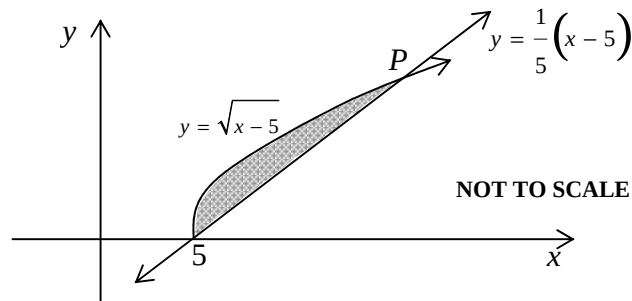
Question 4 (12 marks)

Marks

(a) Solve $\cos 2x^\circ = -\frac{1}{2}$ for $0^\circ \leq x^\circ \leq 360^\circ$

2

(b)



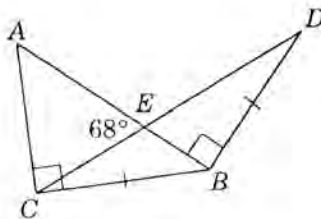
(i) Find the coordinates of P .

2

(ii) Find the area of the shaded region bounded by $y = \sqrt{x-5}$ and $y = \frac{1}{5}(x-5)$.

3

(c)



2

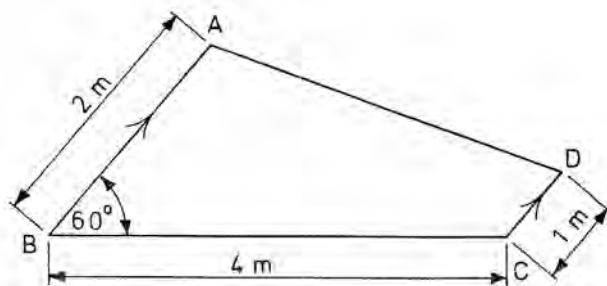
ABC is a right-angled triangle in which $\angle ACB = 90^\circ$.

$\triangle CDB$ is isosceles, in which $CB = DB$.

$\angle AEC = 68^\circ$ and $\angle EBD = 90^\circ$.

Find $\angle DCB$, giving reasons.

(d)



The diagram shows a quadrilateral $ABCD$ with $\angle ABC = 60^\circ$.

$AB = 2$ m, $BC = 4$ m and $DC = 1$ m and $AB \parallel DC$.

(i) Using the cosine rule, calculate AC .

1

(ii) Hence find AD , correct to 3 significant figures

3

SECTION C (Use a SEPARATE writing booklet)

Question 5 (12 marks)

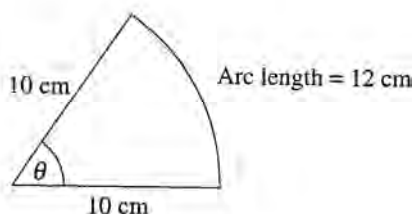
Marks

- (a) A curve $\mathcal{C}$ has equation $y = x^3 - 5x^2 + 7x - 14$.
- (i) Show $\frac{dy}{dx} = (3x - 7)(x - 1)$ 1
- (ii) Find the coordinates of the stationary points and determine their nature. 3
- (iii) Sketch the graph of $\mathcal{C}$, given that an x intercept occurs in the interval $4 \leq x \leq 5$. 2
- (iv) Find the values of x for which $\mathcal{C}$ is concave down. 1
- (b) A polygon has 40 sides.
- The lengths of the sides, starting with the smallest, form an arithmetic series.
- The perimeter of the polygon is 495 cm and the length of the longest side is twice that of the shortest side.
- For this series:
- (i) Find the first term. 3
- (ii) The common difference. 2

Question 6 (12 marks)

Marks

- (a) The diagram below shows a sector of a circle of radius 10 cm. 2
- Find the value of θ to the nearest degree.



- (b) Consider the series $\cos^2 x + \cos^4 x + \cos^6 x + \dots$ for $0 < x < \frac{\pi}{2}$
- (i) Explain why a limiting sum exists. 1
- (ii) Find the limiting sum, expressing the answer in simplest form. 2
- (c) The rate at which people, N , are admitted to Homebake, a music festival in the Domain, is given by
- $$\frac{dN}{dt} = 450t(8-t)$$
- where t is measured in hours.
- (i) Find the maximum rate of people being admitted to the festival. 1
- (ii) If initially $N = 0$, find an expression for the amount of people present at time t . 2
- (iii) The festival *lasted* as long as there was a person there. 1
- How long did the festival last for?
- (d) For the parabola $(y-1)^2 = 16-8x$
- (i) State the coordinates of the vertex and the focus. 2
- (ii) Sketch the graph of the parabola showing the information above 1

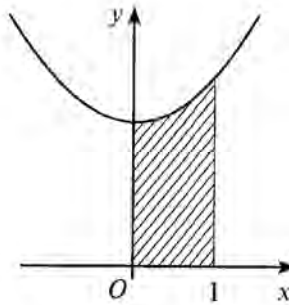
SECTION D (Use a SEPARATE writing booklet)

Question 7 (12 marks)

Marks

(a)

3



The diagram above shows the shaded region bounded by the curve $y = x^2 + 3$, the lines $x = 1$, $x = 0$ and the x axis. This region is rotated 360° about the y axis.

Find the volume generated.

(b)

At time t , the mass M of a material decaying radioactively is given by $M = 5e^{-0.1t}$.

- (i) If at time t_1 , the mass is M_1 and at time t_2 the mass is $\frac{1}{2}M_1$, show that

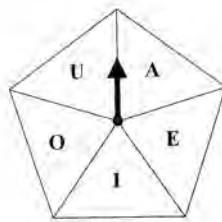
2

$$t_2 - t_1 = 10 \ln 2$$

- (ii) Calculate the time taken for the initial mass to reduce to a mass of $\frac{5}{32}$.

2

(c)



The spinner above is used in a game. *Once spun*, it is equally likely to stop at any one of the letters **A**, **E**, **I**, **O** or **U**.

- (i) If the spinner is spun twice, find the probability that it stops on the same letter twice.
- (ii) How many times must the spinner be spun for it to be 99% certain that it will stop on the letter **E** at least once?

2

3

Question 8 (12 marks)

Marks

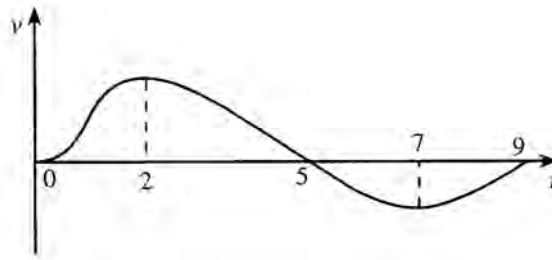
- (a) The velocity v (in km/min) of a train travelling from Olympack Park to Lydcome, non-stop, is given by $v = 20t^2(3-t)$, where t is the time (in minutes) during which the train has been in motion between the two stations.
Find:
- (i) The acceleration of the train at the end of the second minute. 1
 - (ii) Find an expression for the displacement x km of the train from Olympic Park. 2
 - (iii) Hence calculate the distance travelled from Olympic Park to Lidcombe. 1
 - (iv) Where and when, between the two stations, was the train travelling the fastest? 2
- (b) (i) Show $\int_0^1 \frac{dx}{1+x} = \ln 2$ 1
- (ii) By using Simpson's rule with five function values, find an approximation to $\ln 2$. 2
- (c) Yddap is given on his 18th birthday a present of \$500 from his grandparents. 3
- Yddap immediately deposits this into his Credit Union account. His Credit Union gives him a return of 4% pa, compounded annually.
- Each birthday from then on, Yddap decides to deposit \$500 into the same account. He does this up until his 39th birthday.
- His last deposit of \$500 is on his 39th birthday and when Yddap turns 40 he decides to transfer the total of this investment to another account.
- How much does Yddap transfer?

SECTION E (Use a SEPARATE writing booklet)

Question 9 (12 marks)

Marks

(a)



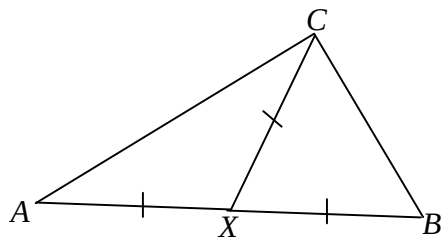
The above graph shows the velocity, $v \text{ ms}^{-1}$, of a particle moving on a straight line, for $0 \leq t \leq 9$.

- | | | |
|--------------|--|---|
| (i) | State all the times, or intervals of time, for which the particle | |
| (α) | is at rest, | 1 |
| (β) | is moving in the positive direction, | 1 |
| (γ) | the acceleration is positive, | 1 |
| (δ) | is slowing down. | 1 |
| (ii) | Using the graph, determine whether the particle has returned to its starting point when $t = 9$. Justify your answer. | 2 |

Question 9 continues on page 11

(b) (i)

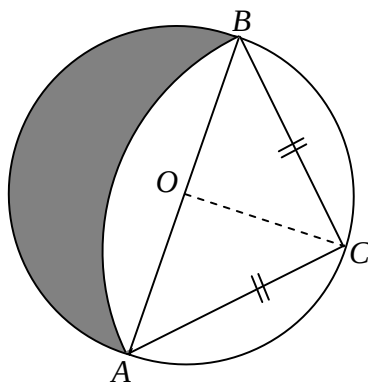
2



The diagram above shows triangle ABC .
 X is a point on AB such that $AX = XB = XC$.

Prove $\angle ACB = 90^\circ$

(ii)



AB is a diameter of the circle ABC whose centre is O .

C is equidistant from A and B .

The arc AB is drawn with C as centre.

(α) If the radius of the circle is r , using (i) show that $AC = r\sqrt{2}$. 1

(β) Hence show that the shaded area is equal to the area of the triangle ABC . 3

Question 10 starts on page 12

SECTION E continued

Question 10 (12 marks)

Marks

A derrick crane is used to lift and move a heavy block across flat ground.

The crane consists of a fixed vertical mast of height m , a boom of fixed length b hinged at the base of the mast, and a hoist rope.

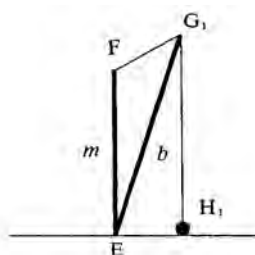


Figure 1

Figure 1 above shows the block is at H_1 on the ground. The hoist rope, anchored at E , passes over pulleys at F and G_1 , then reaches vertically downwards and is attached to the block.

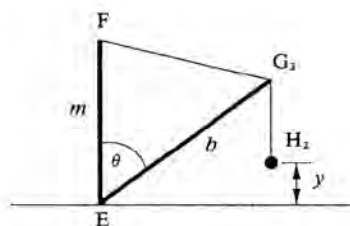


Figure 2

The length of the rope remains constant during the subsequent manoeuvre.

Figure 2 above shows that as the boom is lowered to G_2 , the block moves outwards to H_2 .

Let $\theta = \angle FEG_2$, $0 < \theta < \frac{\pi}{2}$ and let y be the height of the block above the ground.

Assume that $b < 2m$ and that the ground is level. Ignore the size of the pulleys.

- (i) If R is the length of the rope, show that 2

$$y = m + b \cos \theta + \sqrt{b^2 + m^2 - 2bm \cos \theta} - R$$

- (ii) Show that 3

$$\frac{dy}{d\theta} = \frac{bm \sin \theta}{\sqrt{b^2 + m^2 - 2bm \cos \theta}} - b \sin \theta,$$

Question 10 continues on page 13

Question 10 continued

Marks

(iii) Show that when $\frac{dy}{d\theta} = 0$ then either $\cos \theta = \frac{b}{2m}$ or $\theta = 0$. 4

(iv) Assume that y is a maximum when $\cos \theta = \frac{b}{2m}$. 3

The horizontal distance from the mast to the vertical rope is called the *lifting radius* of the crane.

Find the lifting radius in terms of m and b when y is a maximum.

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004

**TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION**

Mathematics

Sample Solutions

Section	Marker
A	Mr Bigelow
B	Mr Hespe
C	Mr Choy
D	Mr Fuller
E	Mr Gainford

QUESTION 1.

(a) $\log(\tan 5/r) = \boxed{1.32.}$ ✓✓

(b) $y = (5x)^{\frac{1}{2}}$
 $y' = \frac{1}{2}(5x)^{-\frac{1}{2}} \times 5$
 $= \boxed{\frac{5}{2\sqrt{5x}}} \checkmark \checkmark \left(\frac{\sqrt{5}}{2} x^{-\frac{1}{2}} \right)$

(c) $2x^2 - x - 15 = 0$

$(2x+5)(x-3) = 0$

$\boxed{x = 3, -\frac{5}{2}}$ ✓✓

(d) $y' = 3 - 2x$
 $\boxed{y = 3x - x^2 + C}$ ✓✓

(e) $y = 2x$ — ①

$3x + 2y = 14$ — ②

Sub ① into ②

$3x + 4x = 14$

$7x = 14$

$x = 2$

Sub in ①

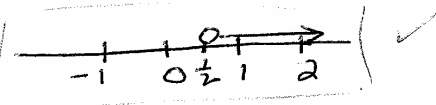
$y = 4$

$\therefore \text{soln. is } \boxed{(2, 4)}$ ✓✓

(f) $3 - 4x < 1$

$-4x < -2$

$\boxed{x > \frac{1}{2}}$ ✓



QUESTION 2.

(a)(i) $f(x) = (1 + \cos 2x)^3$

$$\therefore f'(x) = 3(1 + \cos 2x)^2 \times -2 \sin 2x \\ = \boxed{-6 \sin 2x (1 + \cos 2x)^2} \quad \checkmark \checkmark$$

(ii) $f(x) = x^2 e^{x+2}$

$$f'(x) = x^2 e^{x+2} + 2x e^{x+2} \\ = \boxed{x e^{x+2} (x+2)} \quad \checkmark \checkmark$$

(b)(i) $\int \frac{\cos x}{\sin x} dx = \boxed{\ln(\sin x) + C} \quad \checkmark$

(ii) $\int_{\frac{1}{2}}^2 \left(1 - \frac{1}{x^2}\right) dx = \int_{\frac{1}{2}}^2 (1 - x^{-2}) dx \\ = \left[x + x^{-1} \right]_{\frac{1}{2}}^2 \\ = \left(2 + \frac{1}{2}\right) - \left(\frac{1}{2} + 2\right) \\ = \boxed{0} \quad \checkmark \checkmark$

(c)(i) $\widehat{NBY} = \widehat{MXC}$ (vertical angles)

$\angle$ is common

$\therefore \triangle NBY \cong \triangle MXC$ (equiangular).

$\therefore \boxed{\frac{MX}{NB} = \frac{1}{2}}$ (ratio of corresponding sides equal)

$\left[\frac{NB}{NY} = \frac{1}{2} \text{ (M is the midpt of NY)} \right]$

(ii) $\widehat{MCX} = \widehat{ABC}$ (base angles of an isosceles triangle)

$\widehat{MXC} = \widehat{ABC}$ (vertical angles).

$\therefore \widehat{MCX} = \widehat{MXC}$

$\therefore \triangle XMC$ is isosceles $\therefore MX = MC \therefore \boxed{\frac{MX}{NB} = \frac{MC}{NB} = \frac{1}{2}}$

$$(d) \quad g'(x) = 4 - 3x^2$$

$$\therefore g(x) = 4x - x^3 + C.$$

$$\text{Now } 4 = 4 \times 2 - 8 + C.$$

$$\therefore C = 4$$

$$\therefore \boxed{g(x) = 4x - x^3 + 4} \quad \checkmark \checkmark$$

$$3(i) \quad m_{AD} = \frac{-3-5}{0-5}$$

$$= -8/5 \quad \checkmark$$

$$\theta = \tan^{-1}(-8/5)$$

$$= 180^\circ - 58^\circ$$

$$= 122^\circ \quad \checkmark$$

$$(ii) \quad M = \left(\frac{-5+3}{2}, \frac{1+5}{2} \right)$$

$$= (-1, 3) \quad \checkmark$$

$$(iii) \quad C = (3+(-5-0), 1+(5-3))$$

$$= (-2, 9) \quad \checkmark$$

$$(iv) \quad m_{AB} = \frac{1-3}{3-0}$$

$$= -4/3 \quad \checkmark$$

$$\therefore \text{Eqn of AB is } y = \frac{4x}{3} - 3 \quad \checkmark$$

$$\text{or } 4x - 3y - 9 = 0.$$

$$(v) \quad \text{Distance} = \frac{|4 \times (-5) - 3 \times 5 - 9|}{\sqrt{4^2 + 3^2}}$$

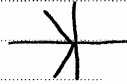
$$= \frac{44}{5} \quad \checkmark \quad (\text{or } 8.8).$$

3 (vi) Length $AB = \sqrt{9+16}$
 $= 5 \checkmark$

$\therefore \text{Area} = \frac{4}{5} \times 5$
 $= 4 \checkmark$

(vii) $x+2y < 5 \checkmark \cap x > 0 \checkmark \cap 4x-3y < 9 \checkmark$

4(a) $2x^\circ = 120^\circ, 240^\circ, 480^\circ, 600^\circ$
 $x^\circ = 60^\circ, 120^\circ, 240^\circ, 300^\circ \checkmark$



(b)(i) $\sqrt{x-5} = \frac{x-5}{5}$

$25x - 125 = x^2 - 10x + 25$

$x^2 - 35x + 150 = 0$

$(x-5)(x-30) = 0$

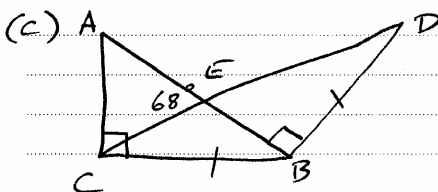
$x = 5, 30 \checkmark$

So P is $(30, 5) \checkmark$

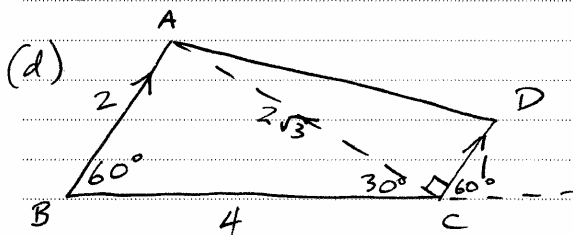
(ii) Area $= \int_5^{30} \left\{ (x-5)^{1/2} - \left(\frac{x-5}{5} \right) \right\} dx \checkmark$

$= \left[\frac{2}{3} (x-5)^{3/2} - \frac{x^2}{10} + x \right]_5^{30} \checkmark$

$= \frac{2}{3} (125 - 0) - \left(\frac{900}{10} - \frac{25}{10} \right) + 30 - 5$
 $= \frac{125}{6} \checkmark \text{ or } 20 \frac{5}{6} \text{ or } 20.8\bar{3}$



$\hat{BED} = 68^\circ$ (vert. opp. $\angle$ s) $\checkmark$
 $\hat{BDE} = 22^\circ$ (angle sum of Δ) $\checkmark$
 $\hat{DCB} = 22^\circ$ (base angle of isosceles Δ) $\checkmark$



(i) $AC^2 = 4^2 + 2^2 - 2 \times 4 \times 2 \times \cos 60^\circ \text{ m}^2$
 $= 16 + 4 - 16 \times \frac{1}{2} \text{ m}^2$
 $= 12 \text{ m}^2$

$AC = 2\sqrt{3} \text{ m.} \checkmark$

$(3.464101615 \dots)$

$$4(d)(ii) \quad \frac{\sin \hat{BCA}}{2} = \frac{\sin 60^\circ}{2\sqrt{3}}$$

$$\sin \hat{BCA} = \frac{1}{2}$$

$$\hat{BCA} = 30^\circ \quad \checkmark$$

$$\begin{aligned} \hat{ACD} &= 180^\circ - 60^\circ - 30^\circ \\ &= 90^\circ \quad \checkmark \end{aligned}$$

$$\therefore AD^2 = 12 + 1$$

$$\begin{aligned} \therefore AD &= \sqrt{13} \quad \checkmark (3.605551275 \text{ calculator}) \\ &= 3.61 \text{ m} \quad (3 \text{ sig. fig.}) \end{aligned}$$

5 $y = x^3 - 5x^2 + 7x - 14$

(a) (i) $\frac{dy}{dx} = 3x^2 - 10x + 7$ [1]

1 $= (3x - 7)(x - 1)$

(ii) $\frac{dy}{dx} = 0, x = \frac{7}{3}, 1$

When $x = 1, y = -11$ [1]

$x = \frac{7}{3}, y = -12\frac{5}{27} (-12.185)$ [1]

3 $\frac{d^2y}{dx^2} = 6x - 10$

$f''(1) = -4 < 0$

$f''(\frac{7}{3}) = 4 > 0$

$(1, -11)$ max [1]

$(\frac{7}{3}, -12\frac{5}{27})$ min



2

1

(iv) $f''(x) < 0, 6x - 10 < 0, \begin{cases} 6x < 10 \\ x < 5/3 \end{cases}$

(b) $a, a+d, a+2d, \dots, a+39d$

$S_{40} = \frac{40}{2} [2a + 39d]$

$\therefore 495 = 20(2a + 39d)$

$99 = 4(2a + 39d)$ — (1)

$a + 39d = 2a$ — (2)

$a = 39d \therefore d = \frac{a}{39}$ — (3)

Subst (3) into (1)

$99 = 4(2a + 39 \times \frac{a}{39})$

$\therefore 99 = 4(3a) \Rightarrow a = \frac{33}{4} (8.25)$

$\Rightarrow d = \frac{11}{52} \text{ cm } (0.212 \text{ cm})$

2

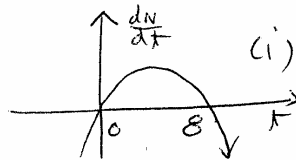
$\frac{495}{20} = 24.75$

$24.75 - 24 = 0.75$

Q (6). (a) $l = 10\theta$
 $\therefore 12 = 10\theta$
 $\theta = 1.2^\circ$
 $l^c = \frac{180}{\pi} \therefore 1.2^\circ = \frac{180 \times 1.2}{\pi} \approx 69^\circ$

(b) (i) $r = \cos^2 x$ (< 1).
 (ii) $S = \frac{\cos^2 x}{1 - \cos^2 x} = \frac{\cos^2 x}{\sin^2 x} = \cot^2 x$

(c) $\frac{dN}{dt} = 450t(8-t)$



(i) $\frac{dN}{dt} (\max)$
 $= 450 \times 4 \times 4$
 $= 7200$

$\frac{dN}{dt} = 3600t - 450t^2$

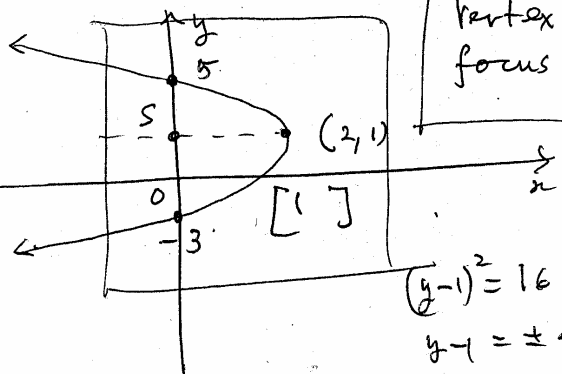
$N = 1800t^2 - 150t^3 + C$ ✓

When $t=0$, $N=0$, $\Rightarrow C=0$

(ii) $\therefore N = 150t^2(12-t)$

(iii) $N=0$, when $t=0$, or 12 .
 $\therefore$ festival last for 12 hrs.

(d) $(y-1)^2 = -8(x-2)$
 $4a = 8$, $a = 2$



vertex $(2, 1)$ [1]
 focus $(0, 1)$ [1]

$(y-1)^2 = 16$
 $y-1 = \pm 4$

3

SECTION D

Question 7

(a) when $x=0$ $x=1$
 $y=3$ $y=4$

$$\begin{aligned} V &= \pi r^2 h - \pi \int_3^4 x^2 dy \\ &= \pi(1)^2(4) - \pi \int_3^4 (y-3) dy \\ &= 4\pi - \pi \left[\frac{1}{2} y^2 - 3y \right]_3^4 \\ &= 4\pi - \pi \left\{ \left[\frac{1}{2}(4)^2 - 3(4) \right] - \left[\frac{1}{2}(3)^2 - 3(3) \right] \right\} \\ &= 4\pi - \frac{\pi}{2} \\ &= \frac{7\pi}{2} \text{ units}^3 \end{aligned}$$

(b)(i) when $t = t_1$, $M = M_1$, ie $M_1 = 5e^{-0.1t_1}$
 when $t = t_2$, $M = \frac{1}{2}M_1$, ie $\frac{1}{2}M_1 = 5e^{-0.1t_2}$
 $M_1 = 10e^{-0.1t_2}$

$$\begin{aligned} 5e^{-0.1t_1} &= 10e^{-0.1t_2} \\ e^{-0.1t_1} &= 2e^{-0.1t_2} \\ -0.1t_1 &= \ln(2e^{-0.1t_2}) \\ -0.1t_1 &= \ln(2) + \ln(e^{-0.1t_2}) \\ -0.1t_1 &= \ln(2) + -0.1t_2 \\ t_1 &= -10\ln(2) + t_2 \\ t_2 - t_1 &= 10\ln(2) \end{aligned}$$

(ii) $\frac{5}{32} = 5e^{-0.1t}$
 $e^{-0.1t} = \frac{1}{32}$
 $-0.1t = \ln\left(\frac{1}{32}\right)$
 $t = -10\ln\left(\frac{1}{32}\right)$
 $t = 34.657$ correct to 3 decimal places

(c)(i) $P(\text{same letter twice}) = 1 \times \frac{1}{5}$

(ii) $P(E \text{ at least once}) = 1 - P(\text{no } E)$

$$= 1 - \left(\frac{4}{5}\right)^n$$

$$\begin{aligned} P(E \text{ at least once}) &\geq \frac{99}{100} \\ 1 - \left(\frac{4}{5}\right)^n &\geq \frac{99}{100} \\ \left(\frac{4}{5}\right)^n &\leq \frac{1}{100} \\ n \ln\left(\frac{4}{5}\right) &\leq \ln\left(\frac{1}{100}\right) \\ n &\geq \frac{\ln\left(\frac{1}{100}\right)}{\ln\left(\frac{4}{5}\right)} \\ n &\geq 20.6377 \dots \\ \therefore n &= 21 \end{aligned}$$

Question 8

(a) $v = 20t^2(3-t)$ $v = 60t^2 - 20t^3$

(i) $a = \frac{dv}{dt}$
 $a = \frac{d(60t^2 - 20t^3)}{dt}$
 $a = 120t - 60t^2$
 when $a = 2$, $a = 120(2) - 60(2)^2$
 $a = 0$

(ii) $x = \int \frac{dx}{dt} dt$
 $x = 20t^3 - 5t^4 + C$
Note: $t = 0, x = 0 \therefore C = 0$
 Hence, $x = 20t^3 - 5t^4$

(iii) Let $v = 0$.
 $0 = 20t^2(3-t)$
 $t = 0$ or $t = 3$

when $t = 3$, $x = 20(3)^3 - 5(3)^4$

Therefore the distance travelled from station to station is 135km.

$$a = 500(1.04)$$

$n = 22$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_{22} = \frac{500 \times 1.04(1.04^{22} - 1)}{1.04 - 1}$$

$$\begin{aligned} \text{(b)(i)} \quad \int_0^1 \frac{dx}{1+x} &= [\ln(1+x)]_0^1 \\ &= \ln(1+(1)) - \ln(1+(0)) \\ &= \ln(2) - \ln(1) \\ &= \ln(2) \end{aligned}$$

$$\int_a^b f(x)dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$\int_0^1 \frac{dx}{1+x} = \int_0^{0.5} \frac{dx}{1+x} + \int_{0.5}^1 \frac{dx}{1+x}$$

(c)

matures to $500(1.04)^{22}$ matures to $500(1.04)^{21}$ matures to $500(1.04)^{20}$

↓

↓

matures to $500(1.04)$

Yddap transferrs

SECTION E

Question 9

- (a) (i) (d) At rest $t=0$
 $t=5$
 $t=9$

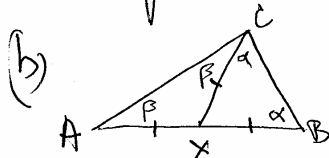
(b) $0 < t < 5$ [1]

(c) $0 < t < 2, 7 < t < 9$ [1]

* (e) $2 < t < 5, 7 < t < 9$ [1]

(ii) No, it has not returned.

The area under the curve is the measure of distance travelled. The negative area (5, 9) is less than the positive area (0, 5).



Aim To prove $\angle ACB = 90^\circ$

Proof let $\angle ABC = \alpha$
 $\angle BAC = \beta$

Now $\angle BCX = \alpha$ (Isos Δ)

and $\angle ACX = \beta$ (")

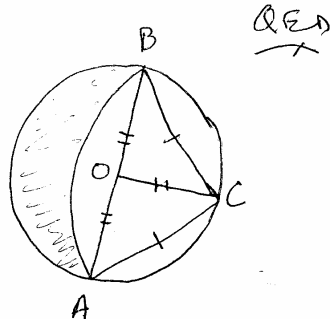
But $\angle BAC + \angle ACB + \angle CBA = 180^\circ$
 (angle sum)

$$\therefore \beta + (\beta + \alpha) + \alpha = 180^\circ$$

$$2(\alpha + \beta) = 180^\circ$$

$$\alpha + \beta = 90^\circ$$

$$\therefore \angle ACB = 90^\circ$$
 [2]



(d) In ΔABC , $AO = BO = CO$
 (radii)
 $\therefore \angle ACB = 90^\circ$

Since $AO = r$, in ΔABC , using Pythagoras

$$AB^2 = 2AC^2$$

$$(2r)^2 = 2AC^2$$

$$4r^2 = 2AC^2$$

$$AC^2 = 2r^2$$

$$\therefore AC = r\sqrt{2}$$
 [1]

(b) $\Delta_{ABC} = \frac{1}{2} (r\sqrt{2})^2$
 $= r^2$

$$A_{\text{shaded}} = \frac{1}{2} \pi r^2 - A_{\text{segment}}$$

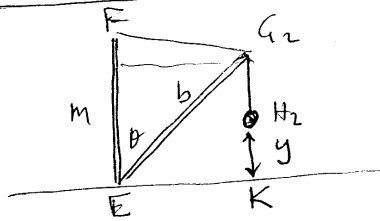
$$= \frac{1}{2} \pi r^2 - \frac{1}{2} (r\sqrt{2})^2 \left(\frac{\pi}{2} - \sin \frac{\pi}{2} \right)$$

$$= \frac{1}{2} \pi r^2 - r^2 \left(\frac{\pi}{2} - 1 \right)$$

$$= r^2$$

$$= \Delta_{ABC}$$
 QED

Question 10



$$\begin{aligned} \text{(i)} \quad R &= FG_2 + G_2K - y \\ \therefore y &= FG_2 + G_2K - R \\ &= \sqrt{b^2 + m^2 - 2bm \cos \theta} + b \cos \theta - R \\ &\text{as required.} \quad [2] \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \frac{dy}{d\theta} &= -b \sin \theta + \frac{1 \times 2bm \sin \theta}{2\sqrt{b^2 + m^2 - 2bm \cos \theta}} \\ &= \frac{bm \sin \theta}{\sqrt{b^2 + m^2 - 2bm \cos \theta}} - b \sin \theta \quad [3] \end{aligned}$$

$$\text{(iii)} \quad \text{When } \frac{dy}{d\theta} = 0$$

$$0 = \frac{bm \sin \theta - b \sin \theta \sqrt{b^2 + m^2 - 2bm \cos \theta}}{\sqrt{b^2 + m^2 - 2bm \cos \theta}}$$

$$\text{i.e. } 0 = b \sin \theta (m - \sqrt{b^2 + m^2 - 2bm \cos \theta})$$

$$\therefore \text{Either } \sin \theta = 0 \quad \text{i.e. } \theta = 0$$

$$\text{or } m - \sqrt{b^2 + m^2 - 2bm \cos \theta} = 0$$

$$\text{i.e. } m = \sqrt{b^2 + m^2 - 2bm \cos \theta}$$

$$\text{That is } b^2 = 2bm \cos \theta$$

$$\text{i.e. } \cos \theta = \frac{b}{2m}$$

[4]

(2)

$$\text{(ii)} \quad [\text{Clearly } y \text{ is min when } \theta = 0]$$

$$\therefore y \text{ is max when } \cos \theta = \frac{b}{2m}$$

Now height radius is $b \sin \theta$

$$= b \sqrt{1 - \cos^2 \theta}$$

$$= b \sqrt{1 - \frac{b^2}{4m^2}}$$

$$= \frac{b}{2m} \sqrt{4m^2 - b^2}$$

[3]